FoilsMode

5D Bayesian Optimization of the Mu2e Stopping-Target Foil Stack

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Mu2e — autoresearch / closed-loop BO

Motivation: why a new BO line?

- 4D helical-plug BO has **saturated**:
  - HV +1.6 % over last 76 evals
  - Hit-rate decayed 62 % → 38 %
- More 4D evals will not buy us much.
- **The next win comes from a dimensionality lift**, not from running the same loop longer.
- Open a parallel BO line over the **stopping-target foil stack** — orthogonal to the helical plug.

The physical knob

Base 37-foil stack (deployed v02):

- `rOut = 75 mm`, `halfThickness = 0.0528 mm` (≈ 105.6 μm full)
- `holeRadius = 21.5 mm`, 37 foils, fixed spacing
- **Pinned** — not optimized.

The +12 envelope (optimized):

- ≤ 6 extra foils **upstream**
- ≤ 6 extra foils **downstream**
- All extras share **one** (rOut, halfThickness, rIn) triple

Helical plug OFF (`hasTSdA = false`, `tsda.helical.build = false`)
→ keeps result orthogonal to bo-helical.

5D search space

Knob	Type	Range
`n_up`	Integer	0 – 6
`n_down`	Integer	0 – 6
`extra_rOut`	Real	50 – 250 mm
`extra_halfThickness`	Real	0.05 – 1.0 mm
`extra_rIn`	Real	0 – 50 mm

Gotcha: `stoppingTarget.holeRadius` is a single scalar at `StoppingTargetMaker.cc:41` — `extra_rIn` is necessarily a **global** override when extras are present.

`is_buildable` rejects `rIn ≥ BASE_ROUT_MM` (vanishing base annulus).

Objective + pipeline

Scalarized objective: `obj = S/√B − α · calo / POT` ($\alpha = 1 \times 10^5$)

Pipeline (per child, 4 grid stages):

```
propose → preflight (G4 surface check)
       → mubeam → run1b_mubeam → concat → mustops_ce
       → harvest → scan_logs → leaderboard
```

- LangGraph state machine, SqliteSaver checkpointed.
- Outer **closed-loop runner**: q parallel children per round, GP refit between rounds.
- Stop criteria: `--max-rounds`, Pareto-hash convergence, or operator `STOP_CLOSED_LOOP`.

Phase 0: preflight passes the extreme corners

Three extreme-corner configs hand-built and run through G4 surface-check **before any grid submit**:

name	n_up	n_down	rOut	hT	rIn	radii len	Result
`foilsP0_AU`	6	0	250	1.0	50	43	**PASS**
`foilsP0_AD`	0	6	250	1.0	50	43	**PASS**
`foilsP0_AS`	6	6	250	1.0	50	49	**PASS**

All three: `total_hits = 1, baseline = 1, managed = 0` — no `StoppingTargetFoil_*` overlap.

Full 5D box is buildable — no defensive clamp needed on the search space.

Bootstrap: no priors, Sobol seed

- **No mmackenz priors** apply: mmackenz v22–v50 are 7D over different knobs (rIn / halfLength4 / holeRadius / col5) — don't project onto this 5D extras-only space.
- Closed-loop GP picker builds its **own** Optimizer with `n_initial_points = q` on round 0 (otherwise skopt: *"Random evaluations exhausted and no model has been fit"*).
- **Parallel strategy:** `cl_min` (collapse-resistant for mixed Integer + Real dims; skopt `cl_mean` warns about fake-y collapse here).

Closed-loop wiring (mode-generic)

Single-line plumbing — FoilsMode rode on the same infrastructure as helical:

- `graph/state.py:32` — `Literal["helical", "michael", "foils"]`
- `graph/closed_loop.py` — `_import_gp(mode)` dispatches to `gp_predict_<mode>`
- `graph/pipeline_io.py` — `mock_metrics` widened to 5D
- New `gp_predict_foils.py` shim (~50 LOC)

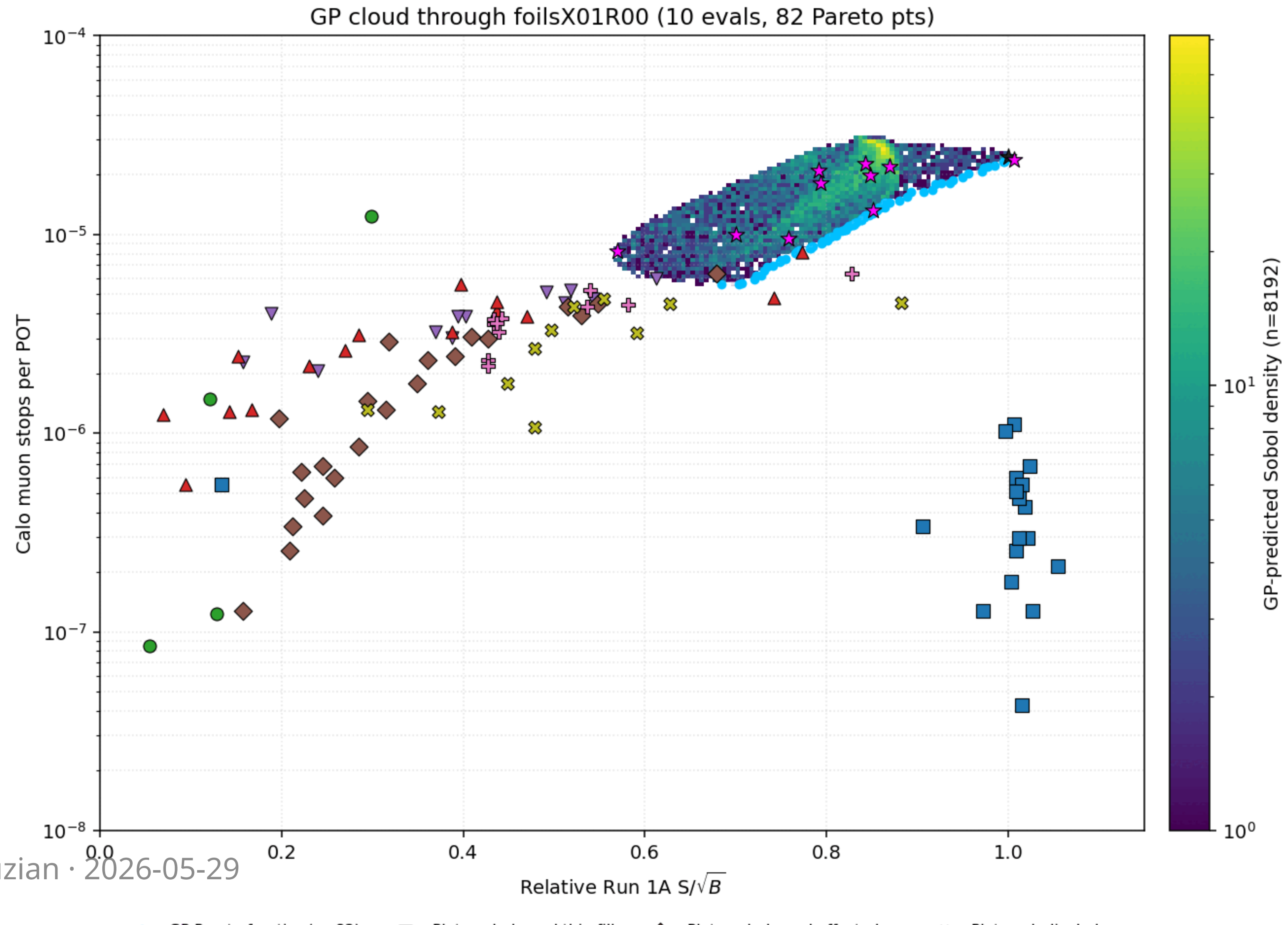
One side-fix: `HelicalMode.FOIL_COUNT = 38 → 37` to match deployed v02 base.

Run timeline

Run	q	Rounds	Evals	Status
X01	10	1	10	Sobol bootstrap, all PASS preflight
X02	10	3	30	clean, frontier still moving
X03	10	5	50	5 distinct Pareto-hashes, non-converged
X04	10	10	100	**in progress** (push past non-convergence)

End of X04: ~190 evals total budget.

GP cloud evolution (animated)



Top configurations so far

config	n_up	n_down	rOut	hT	rIn	sob	calo ($\times 10^{-5}$)	obj
`foilsX03R04_02`	5	6	184.4	0.121	0.0	3.43	1.29	**2.14**
`foilsX03R03_03`	5	6	203.4	0.080	0.0	3.51	1.42	2.09
`foilsX03R04_00`	5	6	174.1	0.125	0.0	3.47	1.40	2.07
`foilsX03R02_06`	5	6	225.4	0.080	0.0	3.41	1.45	1.96
`foilsX03R02_03`	6	6	198.5	0.050	0.0	3.63	1.73	1.90

Pattern (top-5): max-extras dominates — `n_up ∈ {5,6}, n_down = 6, rIn = 0`.
Optimum sits in the **thin-foils / large-radius** regime.

What the GP has learned

Length-scale rails diagnostic (sklearn ConvergenceWarning):

dim	n=10	n=29	n=47
`n_up`	OK	OK	OK
`n_down`	rails	freed	freed
`extra_rOut`	OK	OK	OK
`extra_halfThickness`	OK	OK	OK
`extra_rIn`	rails	rails	**rails**

Take-away: `extra_rIn` remains the worst-trained dim
→ next batch should **explicitly probe rIn** instead of trusting the GP's marginal there.

Bugs hit + fixes (framework hardening)

1. **`numpy.int64` msgpack crash** — skopt `Integer` dims returned `np.int64`; LangGraph SqliteSaver checkpoint died at round transition.
   Fix: cast Integer picks to native `int` in `gp_predict_foils.compute_explore_picks`.
2. **Barrier false-positive on round ≥ 1** (helical-side; inherited to foils) — empty `StateSnapshot` indistinguishable from terminal.
   Fix: require `snap.next` empty **AND** `snap.values` non-empty **AND** `metadata.step ≥ 1`.
3. **Concat convergence-poll never-converges** — `pipeline.py:470` counted only **bare-form** (`00000`) outstage dirs as "settled"; hash-suffix dirs from failed art jobs spun the poll forever.
   Fix: failure-aware exit when `in_queue == 0` AND all `njobs` dirs present in either form.

Lesson: convergence-by-hash is the wrong stop criterion early

Across **X03** (5 rounds, 50 evals):

- Pareto-hashes: `a8f16932 → ddf0adc1 → b30ac643 → 62608fc9 → 6459d20d`
- **All 5 distinct** → no convergence by `k = 2` repeat criterion.
- But frontier still **measurably expanded** (sob 3.52 → 3.87, calo 1e-6 → 8.3e-7).

The hash flips on every new Pareto-dominating point — by design, but useless as a stop criterion in early phase.

Use instead: HV-delta (hyper-volume increment) or absolute eval-budget cap.

Open questions / next steps

- **foilsX04** running now (100 evals, $q = 10 \times 10$ rounds) — push past X03 non-convergence.
- If rIn still rails after X04: **hand-seed** small-rIn / large-rIn probes.
- Re-render **GP cloud at $n \geq 60$** — current dim-4 length-scale rail invalidates extrapolation.
- Consider promoting a **6th dimension** (e.g. base hole radius decoupled from extras) if 5D plateaus.
- **Cross-compare X04 final frontier** with helical Pareto envelope — is the foil DOF additive or partially redundant?

Backup: references

Wiki pages

- bo-foils
- closed-loop-runner
- stopping-target-foil-base-spec

Source files

- `autoresearch_bo_michael.py` — `FoilsMode` class (~L547)
- `graph/closed_loop.py` — multi-round Pareto-pick driver
- `gp_predict_foils.py` — skopt EI picker shim
- `gp_predict_foils_cloud_anim.py` — per-cohort GIF renderer

Leaderboard

`/exp/mu2e/app/users/oksuzian/autoresearch/leaderboard_bo_foils_v1.tsv`